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Does Local Access to Finance Matter? Evidence from U.S. Oil and Natural Gas Shale Booms

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Abstract. I exploit exogenous changes in local credit supply from shale discoveries to identify where and when local access to finance is economically important for firms. I find that the real effects of a credit supply shock are linked to local lending market characteristics. Changes in local credit supply have significant real effects in lending markets dominated by small banks, while there is no effect in other markets. These results suggest that small banks, which do not benefit from deep internal capital markets, face economically meaningful frictions in accessing external interbank markets to fund loans. The effect in other markets, which have large multimarket banks, indicates that these markets are not as dependent on local funding sources for lending. These findings illustrate the important role that the internal capital markets of banks have on real outcomes.

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Keywords: credit supply • bank lending • bank deposits • shale • exogenous shock

1. Introduction

In frictionless financial markets, entrepreneurs and firms should be able to obtain funding for all positive net present value projects. In such a world, changes in local credit supply would have no effect on real outcomes. However, if information or agency frictions interfere with capital mobility, then suboptimal outcomes can occur. A central dimension that influences these frictions is a bank's access to internal capital markets versus external interbank markets for the funding of loans. If banks are large, capital can be redeployed geographically via internal capital markets to fund projects. However, if banks are small, it could be more difficult for capital to be redeployed from other areas to be lent locally, as external financing via interbank funding would need to be available.[1] Despite the importance of these differential funding frictions, direct empirical evidence characterizing the link between bank internal capital markets and real outcomes remains limited.

To identify the role that bank internal capital markets have on real outcomes, I study how lending markets, composed of different types of banks with varying internal capital market depth, respond to localized positive exogenous changes in bank funding. Specifically, I estimate the effect of *similar* changes in local credit supply on real outcomes in *different* lending markets. The intuition of this empirical strategy is that some banking markets, for example markets that have branches of large multimarket banks, are more likely to benefit from the redeployment of capital geographically via bank internal capital markets to be lent locally. Alternatively, other lending markets, dominated by small local banks, are more reliant on external interbank markets for the funding of loans, to the extent that the local deposit base is insufficient to fund all positive net present value (NPV) projects. If markets dominated by small banks experience economically meaningful frictions in accessing external capital markets, then an increase in local deposits, a key funding source, would increase local credit supply and affect local firms.[2] Therefore, the extent to which a positive local deposit shock does affect firms is an indication of the relative role that bank internal capital markets play in facilitating the flow of capital to affect real outcomes.

I use local exogenous bank deposit windfalls from oil and natural gas shale discoveries to examine the effect of changes in credit supply on real outcomes. I identify shale discoveries ("booms") at the county level in the seven major shale-producing U.S. states between 2003 and 2009 using a unique data set of 16,731 individual shale wells. Unexpected technological breakthroughs in shale development have caused energy companies to make high payments to individual mineral owners for the right to develop shale discoveries. I find that the increase in individual mineral wealth associated with shale booms raises local bank deposits by 9.3%. These deposits from newly wealthy mineral owners enhance

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a bank's ability to make new loans, resulting in a positive local credit supply shock.[3]

To trace out how real outcomes are affected after a local credit supply shock, I use a difference-in-differences empirical specification. This specification compares the number of business establishments, my outcome measure, before a boom to after a boom across industries with different external financing requirements.[4] Because both credit supply and credit demand may be changing in a shale boom, I focus on within-county-year comparisons. Specifically, to identify the causal effect of changes in credit supply I include county-year fixed effects, so that any demand effect that impacts industries similarly in a given county in a given year is controlled for.

I find that after a shale-boom–driven credit supply shock, the number of business establishments in industries with high external finance requirements increases 4.6% relative to industries with low external finance requirements.[5] However, this figure varies across different lending markets. I find that the effect of changes in credit supply on local firms is strongly linked with local banking market structure. Areas dominated by small banks, my proxy for markets that have a limited ability to access the deep internal capital markets of multimarket large banks, benefit the most from an expansion in local credit supply. Specifically, after a boom, the number of business establishments in industries with high external finance requirements increases 7.1% relative to the number with low external finance requirements in counties dominated by small banks, whereas there is no change in other lending markets. This result suggests a central role for the dependency of a local lending market on the internal capital markets of its banks and branches, for real outcomes. If internal capital markets and external capital markets were a perfect substitute, one would expect that small banks would have purchased securities with the increased deposits or lent the deposits out in the interbank market. Because local economic outcomes do change in lending markets comprised of small banks, it suggests that the external capital market frictions they face, relative to the internal capital markets that other banks have, adversely affect real outcomes.

Consistent with the growth of establishments in high external finance requirement industries, I document lending increases by small banks in shale boom counties.[6] However, non-credit-based interpretations of my results may still be a concern. For example, some industries could benefit differentially from a shale discovery due to consumer demand shocks, wealth shocks, or other non-credit-based shocks associated with a shale discovery. If any of these shocks are correlated with external financing requirements, then a credit-supply–based interpretation of the results could be problematic. However, for these alternative shocks to alter the interpretation of my empirical design, they would also need to be correlated with the size of a county's local banks. I find no evidence that after booms, demand shocks differ across counties with different bank sizes. Specifically, retail sales, a proxy for local demand, increase by similar amounts after booms in counties dominated by small banks as they do in other counties. Additionally, there is no evidence that deposits increase more after booms in counties dominated by small banks than in other counties, as one might expect if demand shocks affected counties differently. More broadly, the empirical design of this paper requires an alternative, non-banking-based interpretation of results to reconcile why outcomes for industries with distinct external financing requirements respond differently after a shale boom, and why these different responses are larger in counties dominated by small banks.[7]

How are shale booms different than other types of economic growth? I argue that the key differentiator of shale booms is the significant relative increase in local credit supply in shale counties, relative to other types of growth shocks. Because county banking market structure is not randomly assigned, a concern may be that the real outcomes I observe are not driven by a deposit effect, but instead, an omitted variable that affects how certain counties or certain industries respond to economic growth (e.g., rural and underdeveloped areas may respond differently when there is growth). To attempt to identify how this might be influencing my tests, I examine whether nonshale growth shocks affect counties dominated by small banks differently or firms with greater external financing requirements differently. I find no evidence of differential affects linked to county banking market composition or industry external financing requirements in response to nonshale growth shocks. This result is consistent with the credit supply component of shale booms being a key factor for real outcomes, relative to other types of economic growth.

Are banks using shale deposit windfalls to fund positive net present value projects? While difficult to test empirically, there are at least two pieces of suggestive evidence that indicate that banks are not making bad loans. First, in an analysis of banks that have all of their operations in shale counties, for which Call Report data may be considered plausibly representative of the loans a bank may be making in a shale county, I find no evidence that a bank's nonperforming loan ratio increases after a shale boom. Second, establishments in industries with high external finance requirements represent a smaller portion of the economy in lending markets dominated by small banks. Specifically, in nonshale counties dominated by small banks, they comprise 37.8% of all establishments in 2009. In lending markets dominated by small banks that have benefited from a shale boom, this figure is 40.8%. This

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amount is nearly equal to the 40.7% they comprise in lending markets with a greater presence of large banks. Thus, these additional establishments increase only to an amount similar to their proportion in counties with a greater presence of large banks.

One should be cautioned against interpreting the results of this study as suggesting that the existence of small banks is suboptimal. Because of the type of borrowers small banks may serve, and the potential difference in borrowers in counties dominated by small banks relative to other counties, it is not clear that more big banks would improve outcomes. Alternatively, this study does suggest that improved access to funding in areas dominated by small banks does lead to improved outcomes. It implies that the difference in having to obtain funding from an external capital market versus internal capital market has a real effect on economic outcomes. The results would suggest that additional tools or innovations that could mitigate information or agency frictions for small banks in obtaining funding may improve outcomes in areas dominated by small banks.

This study also highlights a bright side, linked to the limited impact of frictions in some lending markets, as areas with a significant presence of large banks are largely unaffected by changes in local credit supply. This suggests that some economically important lending frictions in some places have been mitigated, relative to what prior studies have found (Peek and Rosengren 2000, Becker 2007).

In Section 2, I provide an overview of the hypothesis tested in this study and the related literature. Section 3 provides detail on my identification strategy and background on my natural experiment. Section 4 discusses data and variable definitions. Section 5 discusses my results, and Section 6 concludes the paper.

2. Hypothesis Development and Related Literature

The underlying empirical design of this paper is a dual hypothesis test of two sets of frictions: (1) frictions between borrowers and banks, and (2) frictions between banks and access to funds for lending. Both sets of frictions have to be present for the observed results.

If firms could seamlessly access capital regardless of location, then local credit supply, local banking characteristics, and a local bank's ability to obtain external funds for lending would not matter for local economic outcomes. Any local negative credit shock would be counteracted by distant lenders stepping in to fund positive net present value projects. Recent research suggests that geography and distance currently play less of a role in enhancing informational frictions between borrowers and banks due to improved use of information technology. Berger (2003) documents the rise of internet banking, electronic payment technologies, and credit scoring, while Loutskina and Strahan (2009) document the importance of securitization. These advances would suggest a reduced importance of local access to finance, because borrowers can more easily convey information about themselves to banks that are farther away.

Regulatory-based frictions in the United States have also eroded over time, reducing the importance of distance in lending relationships. Banking deregulation in U.S. states has affected output growth rates (Jayaratne and Strahan 1996), the rate of new incorporations (Black and Strahan 2002), the number of firms and firm-size distribution (Cetorelli and Strahan 2006), and entrepreneurship (Kerr and Nanda 2009). Additionally, Bertrand et al. (2007) document that banking deregulation in France leads to better allocation of bank loans to firms and more restructuring activity. Furthermore, deregulation also facilitates broader internal capital markets.

If distance does aggravate information-based frictions between borrowers and lenders, then local credit supply may matter. In particular, if the cost to overcoming distance-related frictions is prohibitive, as could be the case with "soft"-information borrowers, then local credit supply could be important.[8] In this setting, the frictions that a bank faces in obtaining external funding become important for local economic outcomes. Existing literature suggests that bank internal capital markets are important for a bank's ability to fund loans (Houston et al. 1997, Jayaratne and Morgan 2000, Campello 2002, Gilje et al. 2016). Large multimarket banks have the ability to redeploy capital geographically via internal capital markets to fund loans. Alternatively, small banks have less of a geographic footprint and are more dependent on external interbank markets for the funding of loans, to the extent that local deposit supply is not enough to fund all local positive NPV projects. Several papers document that the size-related frictions small banks face are important for lending. For example, Kashyap and Stein (2000) document that monetary policy influences lending for small banks more than for large banks, whereas Bassett and Brady (2002) document that small banks rely more on deposit funding. Therefore, the extent to which a positive local deposit shock does affect lending is an indication of the frictions small banks face; otherwise, positive NPV loans would have already been funded from interbank sources. Moreover, it is also an indication of the frictions that firms face in obtaining loans outside of their local lending market. More broadly, the extent to which deposit shocks do matter for local outcomes in lending markets with bank branches that have varying degrees of internal capital markets is an

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indication of where frictions in the banking system are having the largest effect.

This paper is also more broadly related to other papers that document the importance of access to finance for economic outcomes in different settings earlier in the United States (Peek and Rosengren 2000, Ashcraft 2005, Butler and Cornaggia 2011, Chava and Purnanandam 2011, Samila and Sorenson 2011, Cornaggia 2013, Adelino et al. 2017) and internationally (Guiso et al. 2004, Khwaja and Mian 2008, Paravisini 2008, Iyer and Peydro 2011, Schnabl 2012, Pascali 2016). Additionally, Plosser (2011) uses shale discoveries as an instrument for bank deposits but focuses on bank capital allocation decisions during financial crises. My contribution differs from these papers in that I trace out the importance bank internal capital markets have on real outcomes.

3. Identification Strategy: Shale Discoveries

3.1. Natural Gas Shale Industry Background

The advent of natural gas shale development is one of the single biggest changes in the U.S. energy landscape in the last 20 years. According to the U.S. Energy Information Agency, in its 2011 Annual Energy Outlook, there are 827 trillion cubic feet (Tcf) of technically recoverable unproved shale gas reserves in the United States; this estimate is a 72% upward revision from the previous year. 827 Tcf of natural gas is enough to fulfill all of the United States' natural gas consumption for 36 years. On an energy equivalent basis, 827 Tcf represents 20 years of total U.S. oil consumption or 42 years of U.S. motor gasoline consumption. As recently as the late 1990s, these reserves were not thought to be economically profitable to develop and represented less than 1% of U.S. natural gas production. However, the development of the first major natural gas shale "play" in the United States, the Barnett Shale in and around Fort Worth, Texas, changed industry notions on the viability of natural gas shale. Continued development of drilling and hydraulic fracturing ("fracking") techniques have enabled even more production efficiencies, and today, shale wells have an extremely low risk of being unproductive (unproductive wells are commonly referred to as "dry holes").

Prior to initiating drilling activities, a firm must first negotiate with a mineral owner to lease the right to develop minerals. Typically, these contracts are comprised of a large upfront "bonus" payment, which is paid whether the well is productive or not, and a royalty percentage based on the value of the gas produced over time. Across the United States, communities have experienced significant fast-paced mineral booms. For example, the New Orleans' *Times-Picayune* (2008) reports the rise of bonus payments in the Haynesville Shale, which increased from a few hundred dollars an acre to $10,000 to $30,000 an acre plus 25% royalty in a matter of a year. An individual who owns one square mile of land (640 acres) and leases out his minerals at $30,000/acre would receive an upfront one-time payment of $19.2 million plus a monthly payment equal to 25% of the value of all of the gas produced on his lease.[9] The significant personal windfalls people have experienced in natural gas shale booms has led to increases in bank deposits in the communities in which they live. Since the first major shale boom in the Barnett (TX), additional booms have occurred in the Woodford (OK), Fayetteville (AR), Haynesville (LA + TX), Marcellus (PA + WV), Bakken (ND), and Eagle Ford (TX).

3.2. Effect of Boom on Deposits

The booms experienced by communities across the United States due to shale discoveries are exogenous to the underlying characteristics of the affected communities (health, education, demographics, etc.). The exogenous factors driving shale development include technological breakthroughs (horizontal drilling/hydraulic fracturing) and larger macroeconomic forces (demand for natural gas and natural gas prices). The first step in my analysis is to quantify the deposit shock in shale boom counties. Specifically, what is the impact of a shale boom on local deposit supply? To do this, I estimate the following regression model.

$$Deposit_{i,t} = \alpha + \beta_1 Boom_{i,t} + Year\ FE_t + County\ FE_i + \varepsilon_{i,t}$$

$Boom_{i,t}$ is a measure of shale activity; in my tests, I use both logarithm of total shale wells and a binary dummy boom variable to measure the shale boom. $Deposit_{i,t}$ is either the logarithm of deposits summed across all branches in county i at time t or the logarithm of deposits per capita summed across all branches in county i at time t. County fixed effects are included to control for time-invariant county effects, and year effects are included to account for time-varying effects; these enter the specification in the form of $Year\ FE_t$ (year fixed effect) and $County\ FE_i$ (county fixed effect). The key variable of interest in this specification is the coefficient β_1, which indicates the change in $Deposit_{i,t}$ attributable to the $Boom_{i,t}$ variable.

3.3. Effect of a Change in Credit Supply on Firms

To identify the economic outcomes related to the local credit supply shock, I use a regression specification that distinguishes between economic outcomes for industries with high external financing requirements relative to those with low external financing requirements. To achieve this aim, I use a regression form of difference-in-differences, where the first difference (β_1) can be thought of as the difference in economic outcomes between boom county-years and nonboom

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county-years. To identify the effect of the credit component of a boom, I incorporate a second difference (β_3), the difference in economic outcomes for industries with high external finance requirements and industries with low external finance requirements.

$$\begin{aligned} &Establishments_{i,j,t} \\ &= \alpha + \beta_1 Boom_{i,t} + \beta_2 High_j + \beta_3 Boom_{i,t} \times High_j \\ &\quad + IndustryYear\ FE_{j,t} + CountyIndustry\ FE_{i,j} \\ &\quad + CountyYear\ FE_{i,t} + \varepsilon_{i,j,t} \end{aligned}$$

where $Establishment_{i,j,t}$ is either the logarithm of the number of establishments in county i and industry group j at time t or the establishments per capita in county i and industry group j at time t. I have grouped establishments into two industry types: one industry group that has high requirements for external finance, for which $High_j = 1$, and one industry group with low requirements for external finance $High_j = 0$.[10] Thus, for every county, I have two industry groups, which are delineated by requirements for external finance. I also include three sets of fixed effects. $IndustryYear\ FE_{j,t}$ control for time-varying differences in industry growth, $CountyIndustry\ FE_{i,j}$ control for county specific differences in industry make-up, and $CountyYear\ FE_{i,t}$ absorbs any county-year specific effects (e.g., demand effects) that might affect firms in both industry groups similarly.

The key variable of interest is the coefficient on the interaction term, β_3. If industries with a high dependence on external finance benefit more from shale booms, β_3 would be positive, which would indicate the importance of the credit supply component of a boom. It is important to note that high and low external finance requirement industries could differ on other, non-credit-related dimensions that could result in a positive β_3, such as differences in local demand. I include a set of tests in the online appendix and in Section 5.5 to assess such alternatives.

3.4. Effect of Bank Size and Credit Supply on Firms

To estimate the importance of local bank size for local credit supply, I use a triple differencing specification. The first two differences are nonboom county-years versus boom county-years and high requirements for external finance versus low requirements for external finance. The third difference tests whether the effect from the first two differences is bigger in areas dominated by small banks: high small-bank market share versus low small-bank market share. This comparison proxies for the extent to which banks rely on external versus internal capital markets. Large multimarket banks can redeploy capital geographically through internal capital markets, while small banks are more dependent on local deposit supply and accessing funds through external capital markets, such as the interbank lending market. $Small\ Bank_{i,t}$ is a variable representing small-bank market share in county i at time t. To measure small-bank market share, $Small\ Bank_{i,t}$, I use both the proportion of branches in a county that belong to small banks as well as a dummy variable for the counties that are above median in small-bank-branch market share in any given year. The interaction of $Small\ Bank_{i,t}$ with the other terms in the specification yields a regression form of difference-in-difference-in-differences.[11]

$$\begin{aligned} &Establishments_{i,j,t} \\ &= \alpha + \beta_1 Boom_{i,t} + \beta_2 High_j + \beta_3 Small\ Bank_{i,t} \\ &\quad + \beta_4 Boom_{i,t} \times High_j + \beta_5 Boom_{i,t} \times Small\ Bank_{i,t} \\ &\quad + \beta_6 High_j \times Small\ Bank_{i,t} + \beta_7 Boom_{i,t} \times Small\ Bank_{i,t} \\ &\quad \times High_j + IndustryTrends\ FE_{j,t} + CountyIndustry\ FE_{i,j} \\ &\quad + CountyYear\ FE_{i,t} + \varepsilon_{i,j,t} \end{aligned}$$

In this regression, the key variable of interest is β_7. If industries with higher requirements for external finance benefit more from a local credit supply shock in counties dominated by small banks, this coefficient would be positive.

4. Data and Variable Definition

For my panel data set, I include the seven states that have experienced shale development activity from 2000 through 2009. These are Arkansas, Louisiana, North Dakota, Oklahoma, Pennsylvania, Texas, and West Virginia. There are 639 counties in these states with at least one bank branch over the sample period. This sample includes counties that have experienced shale booms, as well as counties that have not, and it is these nonboom county-years that serve as a control group in empirical tests. The data are constructed on an annual frequency and compiled from four different sources: (1) well data (from Smith International Inc.), (2) deposit and bank data (from FDIC Summary of Deposits Reports), (3) county-level economic outcome data by industry (Census Bureau, Establishment Data), and (4) external finance requirement measures (from Compustat).

4.1. Well Data

Well data are used to calculate the $Boom_{i,t}$ variables in the regressions. The well data are obtained from Smith International Inc., which provides detailed information on the time (year), place (county), and type (horizontal or vertical) of well drilling activity. I use horizontal wells as the key measure of shale development activity, as the majority of horizontal wells in the United States drilled after 2002 target shale or other unconventional formations. To best measure the influence of shale development activity, I focus on two different measures.

- $Boom_{i,t} = Dummy_{i,t}$: a dummy variable set to 1 if county i at time t is in the top quartile of all county-years with shale well activity (total shale

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Table 1. Summary Statistics of States and Counties With Shale Booms

Panel A: States, counties, shale well activity	
Number of states	7
Number of counties	639
Number of boom counties	104
Total number of shale wells	16,731
Time period	2000–2009

Panel B: Shale discoveries ("booms")

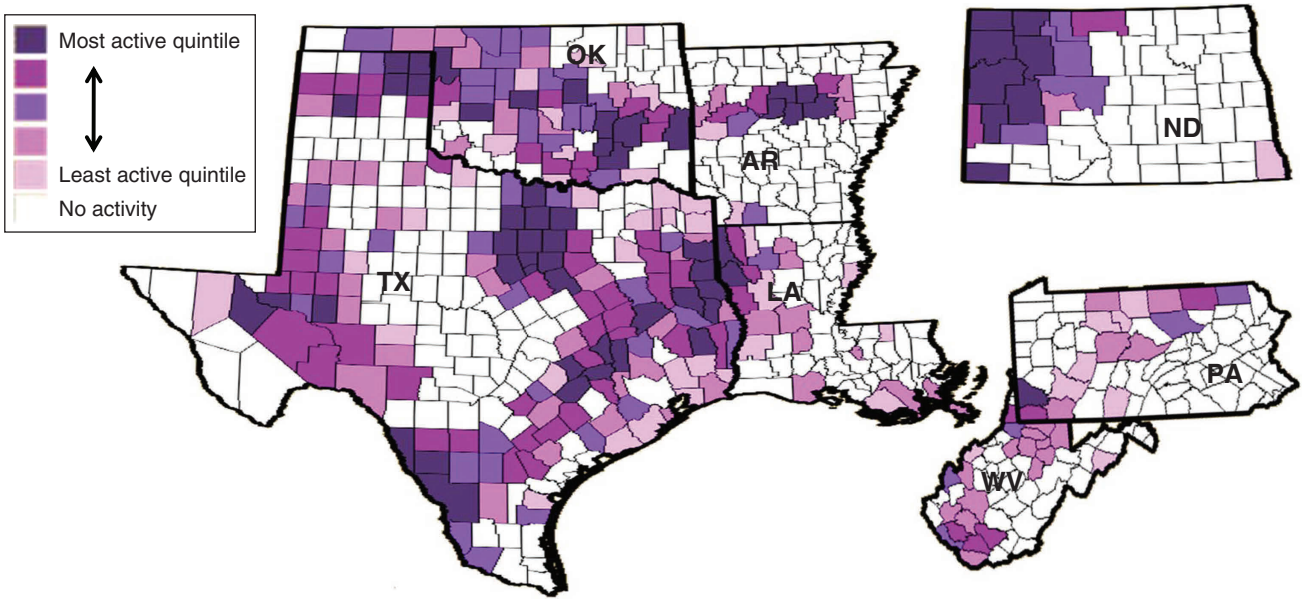
	County	Boom year		County	Boom year
1	Bowman County, North Dakota	2003	53	Bosque County, Texas	2007
2	Brazos County, Texas	2003	54	Ector County, Texas	2007
3	Moore County, Texas	2003	55	Erath County, Texas	2007
4	Potter County, Texas	2003	56	Hill County, Texas	2007
5	Upton County, Texas	2003	57	Jack County, Texas	2007
6	Washington County, Texas	2003	58	Jasper County, Texas	2007
7	Haskell County, Oklahoma	2004	59	Madison County, Texas	2007
8	Pittsburg County, Oklahoma	2004	60	Midland County, Texas	2007
9	Denton County, Texas	2004	61	Panola County, Texas	2007
10	Fayette County, Texas	2004	62	Somervell County, Texas	2007
11	Grimes County, Texas	2004	63	Webb County, Texas	2007
12	Johnson County, Texas	2004	64	Zavala County, Texas	2007
13	Lipscomb County, Texas	2004	65	Cleburne County, Arkansas	2008
14	Maverick County, Texas	2004	66	Atoka County, Oklahoma	2008
15	Shelby County, Texas	2004	67	Latimer County, Oklahoma	2008
16	Terrell County, Texas	2004	68	Lincoln County, Oklahoma	2008
17	Wise County, Texas	2004	69	Roger Mills County, Oklahoma	2008
18	De Soto County, Louisiana	2005	70	Washita County, Oklahoma	2008
19	Billings County, North Dakota	2005	71	Andrews County, Texas	2008
20	McKenzie County, North Dakota	2005	72	De Witt County, Texas	2008
21	Williams County, North Dakota	2005	73	Edwards County, Texas	2008
22	Le Flore County, Oklahoma	2005	74	Ellis County, Texas	2008
23	Gaines County, Texas	2005	75	Freestone County, Texas	2008
24	Hardeman County, Texas	2005	76	Harrison County, Texas	2008
25	Lee County, Texas	2005	77	Hemphill County, Texas	2008
26	Nacogdoches County, Texas	2005	78	Hutchinson County, Texas	2008
27	Parker County, Texas	2005	79	Karnes County, Texas	2008
28	Pecos County, Texas	2005	80	Lavaca County, Texas	2008
29	Reeves County, Texas	2005	81	Live Oak County, Texas	2008
30	Tarrant County, Texas	2005	82	Montague County, Texas	2008
31	Tyler County, Texas	2005	83	Palo Pinto County, Texas	2008
32	Divide County, North Dakota	2006	84	Polk County, Texas	2008
33	Golden Valley County, North Dakota	2006	85	Robertson County, Texas	2008
34	Coal County, Oklahoma	2006	86	Winkler County, Texas	2008
35	Bee County, Texas	2006	87	Logan County, Arkansas	2009
36	Burleson County, Texas	2006	88	Bossier County, Louisiana	2009
37	Dimmit County, Texas	2006	89	Caddo County, Louisiana	2009
38	Hood County, Texas	2006	90	Red River County, Louisiana	2009
39	Houston County, Texas	2006	91	Sabine County, Louisiana	2009
40	Ochiltree County, Texas	2006	92	Bottineau County, North Dakota	2009
41	Roberts County, Texas	2006	93	Canadian County, Oklahoma	2009
42	Ward County, Texas	2006	94	Carter County, Oklahoma	2009
43	Conway County, Arkansas	2007	95	Johnston County, Oklahoma	2009
44	Faulkner County, Arkansas	2007	96	Marshall County, Oklahoma	2009
45	Van Buren County, Arkansas	2007	97	Greene County, Pennsylvania	2009
46	White County, Arkansas	2007	98	Washington County, Pennsylvania	2009
47	Burke County, North Dakota	2007	99	Cherokee County, Texas	2009
48	Dunn County, North Dakota	2007	100	Dallas County, Texas	2009
49	Mountrail County, North Dakota	2007	101	Leon County, Texas	2009
50	Ellis County, Oklahoma	2007	102	San Augustine County, Texas	2009
51	Hughes County, Oklahoma	2007	103	Wheeler County, Texas	2009
52	Oklahoma County, Oklahoma	2007	104	Wood County, Texas	2009

Notes. This table contains summary statistics for the well data used in this study. Development of shale and other unconventional formations is done using horizontal drilling, so I use horizontal well activity as the primary method of measuring when and where booms occur. The states in the sample are states situated in the primary shale development areas: Barnett (TX), Woodford (OK), Haynesville (LA + TX), Fayetteville (AR), Marcellus (PA + WV), Eagle Ford (TX), and Bakken (ND). Well data were obtained from Smith International Inc.

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Figure 1. (Color online) Location and Intensity of Shale Activity



Notes. The figure maps the counties of the seven shale boom states included in this study: OK, TX, LA, WV, PA, ND, and AR. White counties are counties with no shale development activity. The remaining counties are shaded based on intensity of activity related to the total number of shale wells drilled through 2009.

wells > 17) in the panel data set. Once the variable is set to 1, all subsequent years in the panel for the county are set to 1. Based on this definition 88.1% of all shale wells are drilled in boom county-years.

- $Boom_{i,t}$ = Log *Total Shale Wells*$_{i,t}$: the logarithm of the total number of shale wells drilled in county i from 2003 to time t.

Regressions are based on the total shale wells drilled for the year leading up through March. This corresponds to when the County Business Pattern Data are tabulated. Summary statistics on sample states, counties, and well data are presented in Table 1, as well as a detailed list of the shale boom counties used in this study. Figure 1 presents a map of the intensity and location of shale development activity.

4.2. Deposit and Bank Data

Deposit and bank data are obtained from the Federal Deposit Insurance Corporation (FDIC) Summary of Deposit data, which are reported on June 30 of each year and provide bank data for all FDIC-insured institutions. I use the Summary of Deposit data as opposed to data from the Reports of Condition and Income (Call Reports) because Summary of Deposit data provide deposit data at the branch level, while Call Reports only provide data at the bank level. Additionally, Summary of Deposit data provide detailed information on the geographic location of each branch that a bank has, so I can directly observe the branches in boom counties and the banks they belong to. To obtain county-level deposit data, I sum deposits across all branches in a county. To calculate small-bank market share in a county, I calculate the proportion of branches in a county that belong to small banks. I define small banks to be banks with assets below a threshold that could cause a bank to be funding constrained. For the results in this paper, I use $500 million (year 2003 dollars) as the asset threshold for small banks.[12] Prior literature

Table 2. Panel Regression Summary Statistics

	Obs.	Mean	Std. dev.
Deposits			
Log *Deposits*	6,382	12.60	1.41
Deposits ($ in thousands)	6,382	1,195,253	5,228,131
Economic outcomes			
Log *Establishments*	12,764	5.24	1.44
Establishments	12,764	680	2,193
Establishments per Capita (per 10,000 people)	12,764	84.81	34.14
Control/explanatory variables			
Log *Total Shale Wells*	12,764	0.45	1.07
Small-Bank-Branch Market Share	12,764	0.63	0.30

Notes. This table contains summary statistics for the data used in the panel regressions. The unit of observation for economic outcome variables is business establishment counts in a panel data set at the county-year-industry (external finance industry group) level, while the unit of observation for bank deposits is at the county-year level. Data on establishments are from the County Business Patterns survey. Economic outcome variables are summed across all industries into two groups based on an industry's requirements for external finance. Hence, for each county-year, there are two industry groups, one with high requirements for external finance and one with low requirements for external finance. Data on annual population levels are from the Census Bureau. Small banks are categorized as banks with less than $500 million in assets, adjusted for inflation (year 2003 dollars). Bank data were compiled from the FDIC Summary of Deposit reports. Shale well information is based on well data obtained from Smith International Inc.

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(Black and Strahan 2002, Jayaratne and Morgan 2000, Strahan and Weston 1998) has suggested that banks with assets in the $100–$500 million range may be funding constrained. In my empirical tests, I use two measures of small-bank market share. Specifically, I use dummy variables set to 1 for the counties with high small-bank-branch market share (above median) in each year, and 0 otherwise. Additionally, I also use the ratio of small bank branches to total branches in a county. Summary data for bank and branch variables are provided in Table 2.

4.3. County-Level Economic Outcome Data by Industry

Economic outcome data by industry were obtained from the County Business Patterns survey, which is released annually by the Census Bureau. It is worth noting that the survey provides data only at the establishment level, not the firm level—for example, a firm may have many establishments.[13] The survey provides detailed data on establishments and employment in each county, by North American Industry Classification System (NAICS) code as of the week of March 12 every year. My main results are based on economic outcomes grouped at the two-digit NAICS code level, which I match with corresponding Compustat two-digit NAICS code external finance requirement measures. More disaggregated NAICS codes (six-digit NAICS as opposed to two-digit NAICS) provide fewer NAICS code matches to Compustat, which I rely on for external finance requirement measures. I exclude codes 21 (Oil and Gas Extraction), 23 (Construction), 52 (Financials), and 53 (Real Estate) because they may be directly influenced by booms. I exclude 99 (Other) due to lack of comparability with Compustat firms.[14]

After matching County Business Pattern data with Compustat external finance requirement measures, I aggregate all industry codes into two industry groups, one with above-median requirements for external finance (high) and one with below-median requirements for external finance (low). The two-digit NAICS code from the County Business Patterns data is used to obtain an external finance requirement measure from Compustat, which is described in more detail in the next subsection. The objective of the matching is to have the cleanest sorting of NAICS codes into high external finance requirement and low external finance requirement bins. Details on the industries in these bins are provided in Table 3.

4.4. External Finance Requirement Measures

I use an external finance requirement measure similar to the measure used by Rajan and Zingales (1998). The main difference is that while they use this measure only for manufacturing firms, I use it for all industry groups similar to Becker (2007). Specifically, over the 1999–2008 time period for each firm in Compustat, I sum the difference between capital expenditures and operating cash flow. I use the time period 1999–2008 because these fiscal years, which end in December for most public firms, correspond most closely to March of the following year (2000–2009), which is when the county business patterns survey is conducted. By summing over several years, the measure is less susceptible to being driven by short-term economic fluctuations. I then divide this sum by the sum of capital

Table 3. External Finance Requirements of Industries

Two-digit NAICS	Two-digit-NAICS name	External requirement measure	External dependence flag
81	Other Services (except Public Administration)	−0.505	0
42	Wholesale Trade	−0.360	0
62	Health Care and Social Assistance	−0.175	0
44	Retail Trade	−0.127	0
11	Agriculture, Forestry, Fishing and Hunting	−0.079	0
61	Educational Services	−0.012	0
22	Utilities	−0.004	0
48	Transportation and Warehousing	0.071	1
56	Administrative and Support and Waste Management and Remediation Services	0.105	1
72	Accommodation and Food Services	0.183	1
71	Arts, Entertainment, and Recreation	0.418	1
31	Manufacturing	0.475	1
54	Professional, Scientific, and Technical Services	1.023	1
51	Information	1.097	1

Notes. This table reports the industry groups used in this study. The industry groups are based on the two-digit North American Industry Classification System (NAICS) used in the reporting of the County Business Patterns survey, which is reported annually by the Census Bureau. For each industry, a measure of dependence on external finance is calculated, based on the method used by Rajan and Zingales (1998). The external finance requirement measure reported for each industry is the industry median requirement for external finance. The data used to calculate the external finance dependence measure are from Compustat for the period from 1999 to 2008 (the fiscal years that are closest to the March data collection of the County Business Patterns survey from 2000 to 2009). The economic outcome measures used are aggregated into two separate industry groups in each county, one with above-median dependence on external finance (external dependence flag = 1) and one with below-median dependence on external finance (external dependence flag = 0).

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expenditures. Specifically, for firm n, the measure is calculated as

$$ExtFinRequirement_n = \frac{\sum_{1999}^{2008}(CapitalExpenditures_{n,t} - OperatingCashFlow_{n,t})}{\sum_{1999}^{2008} CapitalExpenditures_{n,t}}.$$

I take the median of this measure to get an industry's external finance requirement. The calculation of this measure for each industry is displayed in Table 3.[15]

5. Results

5.1. Effect of Shale Booms on Deposit Levels

Table 4 provides regression results of log deposits and log deposits per capita on different shale boom variables. The evidence suggests a causal relationship between shale booms and bank deposits; specifically, that the individual mineral wealth generated by shale booms translates into more bank deposits. In panel A of Table 4, columns (1) and (2), $Boom_{i,t}$ is found to have both economic and statistical significance. For example, the dummy variable measure of $Boom_{i,t}$ can be interpreted as a boom increasing local deposits by 9.3%. To put this in context, the average annual growth rate in deposits across all counties from 2000 to 2009 was 4.6%, so a boom county would experience an additional increase of 9.3% (4.6% + 9.3% = 13.9% total increase), or a total increase in deposits roughly triple its average annual increase. Figure 2 is a graph of how deposits change over time, and it provides visual evidence that the deposit level does not change substantially until time 0, the first year of the boom.

Further tests will focus on comparisons between counties with high small-bank market share and low

Table 4. Effect of Shale Booms on Bank Deposits

Panel A: Effect of shale booms on bank deposits
$Deposits_{i,t} = \alpha + \beta_1 Boom_{i,t} + Year\ FE_t + County\ FE_i + \varepsilon_{i,t}$

	Dependent variable = Log *Deposits*		Dependent variable = Log *Deposits per Capita*	
	Boom = Dummy (1)	Boom = Log total shale wells (2)	Boom = Dummy (3)	Boom = Log total shale wells (4)
$Boom_{i,t}$	0.093***	0.026***	0.073***	0.018***
	(4.49)	(4.42)	(3.70)	(3.24)
Year FE_t	Yes	Yes	Yes	Yes
County FE_i	Yes	Yes	Yes	Yes
R^2—Within	0.631	0.634	0.604	0.604
N	6,382	6,382	6,382	6,382

Panel B: Effect of shale boom on bank deposits: Counties with different bank sizes
$Deposits_{i,t} = \alpha + \beta_1 Boom_{i,t} + \beta_2 Small\ Bank_{i,t} + \beta_3 Boom_{i,t} \times Small\ Bank_{i,t} + Small\ Bank_{i,t} \times Year\ FE_t + County\ FE_i + \varepsilon_{i,t}$

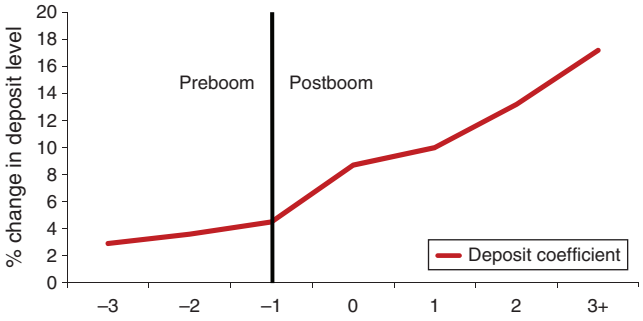
	Dependent variable = Log *Deposits*		Dependent variable = Log *Deposits per Capita*	
	Small bank = Dummy		Small bank = Dummy	
	Boom = Dummy (1)	Boom = Log total shale wells (2)	Boom = Dummy (3)	Boom = Log total shale wells (4)
$Boom_{i,t}$	0.127***	0.029***	0.097**	0.020**
	(3.16)	(2.76)	(2.53)	(2.01)
$Small\ Bank_{i,t}$	Absorbed by Small $Bank_{i,t} \times$ Year FE_t			
$Boom_{i,t} \times Small\ Bank_{i,t}$	−0.047	−0.002	−0.037	−0.002
	(−1.13)	(−0.17)	(−0.94)	(−0.20)
Small $Bank_{i,t} \times$ Year FE_t	Yes	Yes	Yes	Yes
County FE_i	Yes	Yes	Yes	Yes
R^2—Within	0.641	0.643	0.608	0.607
N	6,382	6,382	6,382	6,382

Notes. This table reports the results of regressions that measure the effect of different boom variables on deposits. The dependent variables in these regressions is the log of total deposits in county i in year t and the log of total deposits per capita in county i year t. The explanatory variables are different shale boom variables, which have previously been defined. County and year fixed effects are included to control for time effects and time invariant county effects. Panel A documents the effect of a shale boom on county bank deposits, while panel B tests whether there is a differential effect on bank deposits based on local bank size. The small bank dummy variable used in panel B is equal to 1 if a county has above-median small-bank-branch market share in a given year, and 0 otherwise. Panel B includes the interacted fixed effects of $Small\ Bank_{i,t} \times Year\ FE_t$ to control for differing trends in deposits across counties with different banking market structures. The definition of small bank in these regressions is any bank with less than $500 million in assets adjusted for inflation. Standard errors are clustered by county, with t-statistics reported in parentheses below coefficient estimates.

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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Figure 2. (Color online) Deposit Levels Before and After Shale Boom



Notes. This figure plots the regression of dummy variables based on the year relative to a boom. The first year of a boom is year 0, and the definition of boom that is used is boom dummy (previously defined). For example, the first point is the plot of a dummy variable for time $t-3$ relative to the boom. Due to limited observations for times greater than $t+3$, all observations after time $t+3$ are grouped with the $t+3$ dummy (3+). The dependent variable is the logarithm of total deposits in the county, so the coefficients can be interpreted as the percentage change in the level of deposits at different points in time relative to the boom. Year fixed effects and county fixed effects were included in the regression as well.

small-bank market share. An assumption in this comparison is that both types of counties experience similar deposit shocks. To directly test this assumption I estimate interactions of county bank size characteristics interacted with the shale boom variables. Panel B reports the results of this specification. The key coefficient of interest in assessing whether counties experience different shocks based on their banking structure is the coefficient on the interaction term (β_3). This coefficient is neither economically nor statistically significant, suggesting that counties with different banking structures receive similar deposit shocks.[16]

5.2. Effect of Shale Booms on Lending

Table 5 provides regression results of log loans on different shale boom variables. Individual loan data are typically unavailable in the United States, but there are some data sources that can be used to assess whether there is evidence consistent with a rise in lending in shale boom areas, and in particular areas dominated by small banks. Panel A reports regression results of the amount of commercial and industrial (C & I) loans made by banks that have all of their branches in a single county, and compares lending by single county banks in boom counties versus single county banks in nonboom counties. The regression results can be interpreted as single county banks increasing lending by 3.4% for a one standard deviation increase in log shale wells. Note that this regression specification does not control for demand effects, but it is nonetheless consistent with an increase in lending. Additionally, lending driven purely by demand would be more likely to result in higher interest rates and interest income, whereas Table 5 documents that overall interest income and interest paid on deposits are unchanged after a boom.

Panel B of Table 5 reports regression results of the effect of shale booms on small business lending based on data from the Community Reinvestment Act (CRA). These data identify lending amounts at the county level made by banks to small businesses, but many small banks are not required to report these data, though some do. The results in this test are identified based on those small banks that decide to report their lending and is therefore not comprehensive. Consistent with panel A, this test documents a rise in lending by small banks linked with shale, particularly in areas dominated by small banks.

5.3. Effect of Credit Supply Shock on Firms

To estimate the credit supply effect of a boom, I compare firms in different industries and include county-year fixed effects in regression specifications so that any direct demand effect that both industry groups experience is fully absorbed. Table 6 provides an estimate of the effect of the credit supply shock on firms, using a regression form of difference-in-differences. The coefficient on the interaction term is positive and statistically significant in all specifications, suggesting that firms in industries with high external finance requirements benefit more than firms in industries with low external finance requirements. The economic interpretation of the interaction coefficient in column (1) of Table 6 is that, when there is a boom, establishments in industries with high requirements for external finance increase 4.6% relative to establishments in industries with low requirements for external finance. To put this number in context, the average annual increase in establishments of firms in industries with high external finance requirements from 2000 to 2009 is 0.9%. The interpretation of column (3) in Table 6 is that there are 3.6 additional establishments per 10,000 people after the credit supply shock in industries with high external finance requirements relative to industries with low external finance requirements. While the results in Table 6 could be attributed to a number of potential differences across high and low external finance requirement industries, different local demand for goods and services across industries for example, I provide additional tests that point toward a credit-based explanation of my overall results (see Section 5.5).

5.4. Internal Capital Markets and Credit Supply Shocks

Counties dominated by small banks may benefit more from a credit supply shock if the frictions they face in obtaining finance through external capital markets is greater than the frictions large banks face through there internal capital markets. These frictions could

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Table 5. Bank Lending and Shale Booms

Panel A

$$Bank\ Outcome_{i,j,t} = a + \beta_1 Boom_{i,t} + Year\ FE_t + Bank\ FE_j + \varepsilon_{i,j,t}$$

	Boom = Log total shale wells				
	Outcome = Log deposits	Outcome = Log C&I loans	Outcome = Interest rate	Outcome = Deposit interest rate	Outcome = Non-performing loan ratio
	(1)	(2)	(3)	(4)	(5)
$Boom_{i,t}$	0.0277***	0.0316***	−0.0007	0.0000	−0.0004**
	(4.37)	(2.69)	(−0.88)	(0.36)	(−2.03)
Year FE_t	Yes	Yes	Yes	Yes	Yes
Bank FE_j	Yes	Yes	Yes	Yes	Yes
R^2—Within	0.376	0.103	0.399	0.871	0.025
N	8,176	8,176	8,176	8,176	8,176

Panel B

$$\text{Log } CRA\ Lending_{i,j,t} = \alpha + \beta_1 Boom_{i,t} + Bank\ County_{i,j} + Bank\ Year\ FE_{j,t} + \varepsilon_{i,j,t}$$

	Boom = Log total shale wells			
	High small-bank market share	High small-bank market share	Low small-bank market share	Low small-bank market share
	Large bank	Small bank	Large bank	Small bank
$Boom_{i,t}$	−0.002	0.141*	0.012	0.023
	(−0.11)	(1.73)	(1.15)	(0.47)
Bank County $FE_{i,j}$	Yes	Yes	Yes	Yes
Bank Year $FE_{j,t}$	Yes	Yes	Yes	Yes
R^2—Within	0.850	0.939	0.897	0.930
N	50,019	5,274	97,459	12,087

Notes. This table reports the results of regressions that estimate the effect of different shale boom variables on bank outcomes for banks that have all of their branches in a single county. The unit of observation in this panel is county i, bank j, year t. In panel A, data on banks were compiled from Call Reports and Summary of Deposit reports. A bank is in the sample if all of its branches are in a single county in given year, treatment banks are those banks that are in shale-boom county-years, while control banks are single county banks in non-shale-boom county-years. C&I loans are the total amount of commercial and industrial loans a bank reports on its Call Report. Interest income is the total interest income a bank generates in a year, divided by its average total loans. Deposit interest rate is the interest paid on all deposits divided by the average amount of deposits a bank has in a given year. Both interest rate and deposit interest rate variables were winsorized at 1% and 99%. Year fixed effects and bank fixed effects are included. Standard errors are clustered by bank in panel A. In panel B, the lending data are based on lending data reported by the Community Reinvestment Act (CRA), where the unit of observation is lending in county i, by bank j, at time t. Many small banks are not required to report lending due to their size, but some still elect to, and it is these small banks whose lending is reported in the sample. Bank-year fixed effects are included in panel B to control for any shocks that affected the bank in a given year, while bank-county fixed effects are included to control for the level of lending by the bank in a given year. Clustering in panel B is done at the county level. t-Statistics reported in parentheses below coefficient estimates.

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

be driven, for example, by information and agency frictions in the banking system. To test this in a difference-in-differences framework, I subdivide counties into high small-bank market share and low small-bank market share counties, based on whether a county is above median in small-bank market share in a given year. I estimate the specification presented in Table 6 for each of these subgroups and report the results in Table 7.

In every specification, the counties dominated by small banks have a higher coefficient for the interaction term $Boom_{i,t} \times High_j$. The magnitude of the difference is often quite large, with counties with high small-bank market share (bank = high small-bank market share) having coefficients four to five times higher than the coefficients of counties with low small-bank market share (bank = low small-bank market share), depending on the specification. The interaction coefficient for lending markets with low small-bank market share is often not statistically significant. The economic interpretation of (1) is that establishments in industries with high requirements for external finance increase 7.1% relative to establishments in industries with low requirements for external finance after a shale boom, while the economic interpretation of (2) is that establishments in industries with high requirements for external finance increase 1.2% relative to establishments in industries with low requirements for external finance, though this difference is not statistically significant. These results indicate that there is significant cross-sectional variation in the effect of changes in credit supply linked to banking market structure.

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Table 6. Effect of Credit Supply Shock on Firms: Difference-in-Differences (Boom vs. Nonboom, High External Finance Requirements vs. Low External Finance Requirements)

$$Establishments_{i,j,t} = \alpha + \beta_1 Boom_{i,t} + \beta_2 High_j + \beta_3 Boom_{i,t} \times High_j + IndustryYearFE_{j,t} + CountyIndustryFE_{i,j} + CountyYearFE_{i,t} + \varepsilon_{i,j,t}$$

	Dependent variable = Log *Establishments*		Dependent variable = *Establishments per Capita*	
	Boom = Dummy	Boom = Log total shale wells	Boom = Dummy	Boom = Log total shale wells
	(1)	(2)	(3)	(4)
$Boom_{i,t}$	Absorbed by County × Year $FE_{i,t}$			
$High_j$	Absorbed by County × Industry $FE_{i,j}$			
$Boom_{i,t} \times High_j$	0.046***	0.012***	3.629***	0.974***
	(2.67)	(2.61)	(2.71)	(2.78)
Industry × Year $FE_{j,t}$	Yes	Yes	Yes	Yes
County × Industry $FE_{i,j}$	Yes	Yes	Yes	Yes
County × Year $FE_{i,t}$	Yes	Yes	Yes	Yes
R^2—Within	0.008	0.008	0.008	0.009
N	12,764	12,764	12,764	12,764

Notes. This table reports the results of regressions measuring the effect of a credit supply shock on local firms by comparing the change in establishments between two industry groups, one that has high requirements for external finance (ind = high) and one that has low requirements for external finance (ind = low). The dependent variables in these regressions are log establishments and establishments per capita (per 10,000 people). The regression estimates are a regression form of difference-in-differences and test change in establishments before the boom versus after the boom across industries groups with different external finance requirements. County-industry and industry-year (industry trends) fixed effects are included, as well as county-year fixed effects. Standard errors are clustered by county, with t-statistics reported in parentheses below coefficient estimates.

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table 7. Effect of Credit Supply Shock on Firms: Subdivided by Small-Bank Market Share (Boom vs. Nonboom, High External Finance Requirements vs. Low External Finance Requirements)

$$Establishments_{i,j,t} = \alpha + \beta_1 Boom_{i,t} + \beta_2 High_j + \beta_3 Boom_{i,t} \times High_j + IndustryYearFE_{j,t} + CountyIndustryFE_{i,j} + CountyYear_{i,t} + \epsilon_{i,j,t}$$

	Dependent variable = Log *Establishments*				Dependent variable = *Establishments per Capita*			
	Boom = Dummy		Boom = Log total shale wells		Boom = Dummy		Boom = Log total shale wells	
	High small-bank market share	Low small-bank market share	High small-bank market share	Low small-bank market share	High small-bank market share	Low small-bank market share	High small-bank market share	Low small-bank market share
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Boom_{i,t}$	Absorbed by County × Year $FE_{i,t}$							
$High_j$	Absorbed by County × Industry $FE_{i,j}$							
$Boom_{i,t} \times High_j$	0.071***	0.012	0.019***	0.004*	4.905***	1.443	1.453***	0.329
	(3.10)	(1.09)	(2.86)	(1.73)	(3.03)	(1.35)	(3.16)	(1.46)
Industry × Year $FE_{j,t}$	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County × Industry $FE_{i,j}$	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County × Year $FE_{i,t}$	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2—Within	0.014	0.001	0.013	0.003	0.011	0.003	0.013	0.002
N	6,502	6,262	6,502	6,262	6,502	6,262	6,502	6,262

Notes. This table reports a regression form of difference-in-differences for two different county groups, one county group that has high small-bank market share (bank = high small-bank market share) and one county group with low small-bank market share (bank = low small-bank market share). The definition of small bank in these regressions is any bank with less than $500 million in assets adjusted for inflation. For this regression, counties with high small-bank market share are defined to be counties with above-median small-bank-branch market share. The dependent variables in these regressions are log of establishments and establishments per capita (per 10,000 people). The explanatory variables are different shale boom variables, which have previously been defined. Additionally, an interaction between boom variables and the "high" external finance dependence dummy is included, this is the difference-in-differences coefficient of interest (β_3). County-industry and industry-year (industry trends) fixed effects are included, as well as county-year fixed effects. Standard errors are clustered by county, with t-statistics reported in parentheses below coefficient estimates.

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

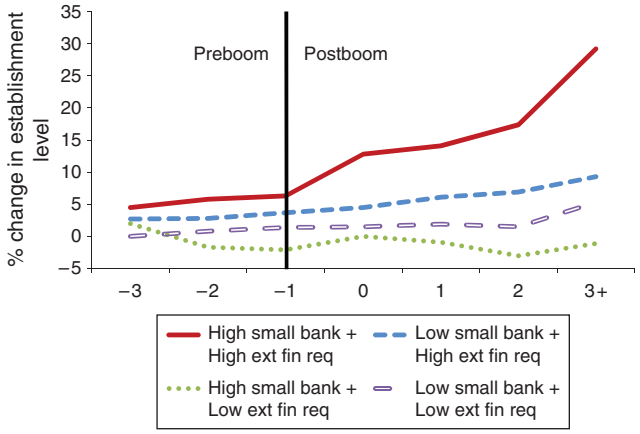
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In the absence of frictions, changes in local credit supply should not affect local firms; because there is a larger effect of changes in credit supply in counties dominated by small banks, it suggests that these lending markets, which do not have access to deep internal capital markets in their banks, are where frictions in the banking system are most problematic. Alternatively, in other lending markets with a greater presence of large banks, there is an economically negligible effect on local firms, which is often not statistically significant. This indicates that these areas have had frictions reduced and benefit from the internal capital markets of large multimarket banks. While non-credit-based explanations—for example, differences in local demand for goods or services across industries or across banking markets—would be compatible with the tests in Tables 6 and 7, I test for a series of demand-based explanations in Online Appendices A and B, and Section 5.5.

I plot how establishments change relative to the boom in Figure 3 and subdivide high external finance and low external finance industries by bank size characteristics to form four separate subgroups. As can be seen, all subgroups trend similarly until time 0, when the subgroup that comprises high external finance requirement industries in counties with high small-bank market share trends higher.[17]

Figure 3. (Color online) Effect of Credit Supply Shock on Counties with Different Bank Sizes



Notes. This figure plots separately the regression coefficients of dummy variables of the year relative to a boom for different subgroups. Specifically, four different group designations are used based on whether an establishment has high or low requirements for external finance and whether it is in a county with high or low small-bank market share. The first year of a boom is year 0, and the definition of boom that is used is boom dummy (previously defined). For example, the first point is the plot of a dummy variable for time $t-3$ relative to the boom. Due to limited observations for times greater than $t+3$, all observations after time $t+3$ are grouped with the $t+3$ dummy (3+). The dependent variable is logarithm of establishments in an industry in a county, so the coefficients can be interpreted as the percentage change in establishment levels at different points in time relative to the boom. Year fixed effects and county fixed effects were included in the regression as well.

To formally test the difference in coefficients across specifications in Table 7 and Figure 3, I estimate a regression form of difference-in-difference-in-differences, with the results shown in Table 8. This is done by adding additional interactions with small-bank market share variables. The coefficient of interest in these tests is the triple interaction term $Boom_{i,t} \times High_j \times Small\ Bank_{i,t}$. The interpretation of (1) in Table 8 is that high external finance requirement establishments increase by 6.2% relative to establishments in industries with low requirements for external finance in boom counties dominated by small banks relative to the difference between these industry groups in other boom counties.[18] Across all specifications, the coefficient on $Boom_{i,t} \times High_j \times Small\ Bank_{i,t}$ is positive and statistically significant, providing evidence suggesting that counties with higher small-bank market share, which do not benefit from deep internal capital markets of banks, are more affected by economically important frictions in the banking system.[19]

The results in Table 8 also address concerns regarding alternative explanations from the prior difference-in-differences tests conducted. For example, it could be the case that high external finance requirement industries benefit more in general when there is an economic boom (high asset beta).[20] However, this explanation would not account for the differential impact experienced in counties with high small-bank market share relative to other lending markets. An additional concern may be that there could be more demand for goods and services for industries in the high external finance dependence industry group. However, for this explanation to be consistent with the results in Table 8, there would also need to be a rationale for why this demand differential is relatively higher in counties with high small-bank market share.

One potential alternative interpretation of Table 8 is that markets dominated by large multimarket banks may export the deposits to connected markets to be lent, as opposed to lending locally in the county of the shale boom. There is evidence that this does occur (Gilje et al. 2016). However, two pieces of evidence indicate that this may not be adversely affecting real outcomes. First, the proportion of high external finance establishments in markets dominated by large banks remains at a similar level to the markets dominated by small banks after shale discovery. One might expect a much higher proportion of high external finance establishments in markets dominated by small banks relative to other areas if these firms were getting underserved in the areas dominated by large banks. Second, I do not observe an increase in lending by small banks in markets dominated by large banks after the shale discovery. One might expect this to occur if credit was

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Table 8. Effect of Bank Size and Credit Supply on Firms: Differences-in-Differences-in-Differences Regression (Boom vs. Nonboom, High External Finance Requirements vs. Low External Finance Requirements, High Small-Bank Market Share vs. Low Small-Bank Market Share)

$$Establishments_{i,j,t} = \alpha + \beta_1 Boom_{i,t} + \beta_2 High_j + \beta_3 Small\ Bank_{i,t} + \beta_4 Boom_{i,t} \times High_j + \beta_5 Boom_{i,t} \times Small\ Bank_{i,t} + \beta_6 High_j \times Small\ Bank_{i,t} + \beta_7 Boom_{i,t} \times Small\ Bank_{i,t} \times High_j + IndustryYear\ FE_{j,t} + CountyIndustry\ FE_{i,j} + CountyIndustry\ FE_{i,t} + \epsilon_{i,j,t}$$

	Dependent variable = Log *Establishments*				Dependent variable = *Establishments per Capita*			
	Boom = Dummy		Boom = Log shale wells		Boom = Dummy		Boom = Log shale wells	
	Bank = Dummy (1)	Bank = Ratio (2)	Bank = Dummy (3)	Bank = Ratio (4)	Bank = Dummy (5)	Bank = Ratio (6)	Bank = Dummy (7)	Bank = Ratio (8)
$Boom_{i,t}$	Absorbed by County × Year $FE_{i,t}$							
$High_j$	Absorbed by County × Industry $FE_{i,j}$							
$Small\ Bank_{i,t}$	Absorbed by County × Year $FE_{i,t}$							
$Boom_{i,t} \times High_j$	0.008	−0.065*	0.003	−0.015*	0.742	−3.079	0.207	−0.937
	(0.49)	(−1.78)	(0.84)	(−1.80)	(0.48)	(−1.05)	(0.59)	(−1.49)
$Boom_{i,t} \times Small\ Bank_{i,t}$	Absorbed by County × Year $FE_{i,t}$							
$Small\ Bank_{i,t} \times High_j$	−0.010	0.023	−0.013	0.011	−0.319	3.803	−0.656	2.767
	(−1.20)	(0.91)	(−1.49)	(0.44)	(−0.37)	(1.32)	(−0.76)	(0.97)
$Boom_{i,t} \times Small\ Bank_{i,t} \times High_j$	0.062**	0.172**	0.015**	0.043***	4.715**	10.378**	1.371***	3.127***
	(2.14)	(2.54)	(2.24)	(2.66)	(2.09)	(2.11)	(2.70)	(2.81)
Industry × Year $FE_{j,t}$	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County × Industry $FE_{i,j}$	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County × Year $FE_{i,t}$	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2—Within	0.013	0.020	0.013	0.021	0.011	0.017	0.015	0.022
N	12,764	12,764	12,764	12,764	12,764	12,764	12,764	12,764

Notes. This table reports results for a regression form of difference-in-difference-in-differences, where the coefficient of interest is the triple interaction term. The dependent variables in these regressions are log establishments or establishments per capita (per 10,000 people) in county i, year t, industry group j. The explanatory variables are different boom variables, which have previously been defined. The definition of small bank in these regressions is any bank with less than $500 million in assets adjusted for inflation. These specifications provide results for two different measures of $Small\ Bank_{i,t}$. One measure is a dummy variable, set to 1 if a county has above-median small-bank-branch market share in any given year and 0 otherwise (Small Bank = Dummy), while the other measure is the ratio of branches that belong to small banks relative to the total number of bank branches in a county (Small Bank = Ratio). Additionally, a set of fully saturated interactions between boom variables, small bank variables, and the high external finance dependence dummy are included. The key coefficient of interest for the difference-in-difference-in-differences regression is the triple interaction term β_7. County-industry and industry-year (industry trends) fixed effects are included, as well as county-year fixed effects. Standard errors are clustered by county, with t-statistics reported in parentheses below coefficient estimates.

*, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

being exported to other geographies at the expense of local firms.

5.5. Validity of Experimental Design

5.5.1. Nonshale Growth Shock. Banking market structure is not randomly assigned; therefore, one concern may be that there are omitted factors that affect both a county's banking market structure and how certain industries (e.g., those with high external finance requirements) are affected by growth shocks. To attempt to assess whether such omitted factors may be affecting my estimates, I conduct a test to assess whether nonshale growth shocks affect one industry group compared to another or one industry group relatively more in counties dominated by small banks. Specifically, in Table 9, I use data from the states immediately adjacent to the seven shale states to test whether nonshale growth shocks or "booms" affect the number of establishments in industries with high external finance requirements differently or the number of establishments in high external finance requirement industries in counties dominated by small banks differently. $Growth\ Shock_{i,t}$ dummy variables are inserted after high-growth county-years so that the number of growth shock county years is approximately the same proportion as the number of shale boom county years obtained in the main sample (5% of all county-years). I obtain growth shock years by identifying years that experience a large increase in the number of business establishments; on average, these growth shocks result in a 17.6% increase in establishments across all industries, a figure significantly higher than shale booms. The key coefficient of interest to test whether industries

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Table 9. Effect of Nonshale Growth Shocks on Firms

	Dependent variable = Log *Establishments*		Dependent variable = *Establishments per Capita*	
	Growth shock = Dummy		Growth shock = Dummy	
	Small bank share = Dummy		Small bank share = Dummy	
	(1)	(2)	(3)	(4)
$Growth\ Shock_{i,t}$	Absorbed by County × Year $FE_{i,t}$			
$High_j$	Absorbed by County × Industry $FE_{i,j}$			
$Small\ Bank_{i,t}$	Absorbed by County × Year $FE_{i,t}$			
$Growth\ Shock_{i,t} \times High_j$	0.002	0.011	−3.591	−3.480
	(0.07)	(0.52)	(−1.54)	(−1.29)
$Growth\ Shock_{i,t} \times Small\ Bank_{i,t}$	Absorbed by County × Year $FE_{i,t}$			
$Small\ Bank_{i,t} \times High_j$		−0.002		−0.468
		(−0.25)		(−0.59)
$Growth\ Shock_{i,t} \times Small\ Bank_{i,t}$		−0.027		−0.338
$\times High_j$		(−0.83)		(−0.10)
Industry × Year $FE_{j,t}$	Yes	Yes	Yes	Yes
County × Industry $FE_{i,j}$	Yes	Yes	Yes	Yes
County × Year $FE_{i,t}$	Yes	Yes	Yes	Yes
R^2—Within	0.000	0.001	0.003	0.003
N	22,932	22,932	22,932	22,932

Notes. This table reports estimates of regressions similar to those presented in Tables 6 and 8, except that the shocks used are nonshale growth shocks in states adjacent to the seven shale states in this study. Specifically, dummy "growth shock" variables are inserted after high-growth county-years such that the number of growth shock county years is approximately the same proportion of shale boom county years obtained in the main sample (roughly 5% of all county-years). The objective of this specification is to test whether general growth shocks differentially affect a particular industry group or a particular set of industries in counties dominated by small banks. The dependent variables in these regressions are log establishments or establishments per capita (per 10,000 people) in county i, year t, industry group j. Standard errors are clustered by county, with t-statistics reported in parentheses below coefficient estimates.

with high external finance requirements are affected differentially by these growth shocks is on the interaction term $Growth\ Shock_{i,t} \times High_j$; this coefficient is not statistically significant. Additionally, the triple interaction term $Growth\ Shock_{i,t} \times High_j \times Small\ Bank_{i,t}$ is neither positive nor statistically significant. These results indicate that industries with high external finance requirements and industries with high external finance requirements in counties dominated by small banks are not differentially affected by general economic growth.

The primary difference between the growth shocks identified in Table 9 and the shale growth shocks used in this study is the relative importance of the credit supply component of the growth shock. Specifically, in a shale boom, overall establishments increase by 2.2% with significant variation linked to external finance requirements (documented in Tables 6 and 8); bank deposits increase by 9.3%, more than four times the overall establishment increase. Alternatively, in the nonshale growth shocks, establishments increase overall by 17.6%, while deposits increase by slightly less than half this amount, a deposit change of less than half the establishment increase compared to the more than four times relative increase in shale booms. These results suggest that the credit component of shale booms make shale growth shocks unique from general localized growth shocks.

The test in Table 9 also provides evidence toward potential concerns regarding how local the demand might be for the products and services produced by different industries. For example, high external finance requirement firms may produce more goods that are sold locally relative to low external finance requirement firms. This potentially could also vary with banking market structure. Such a difference could alter the interpretations of the main tests in Tables 6 and 8. However, if there were substantial differences in local demand for the goods or services of different firms, one would likely expect that a positive growth shock, as outlined in Table 9, would disproportionately affect firms that sell locally or markets where demand for goods and services were likely to be local, since the growth shocks used are local. The results in Table 9 provide some evidence toward there not being substantial differences in local demand of goods or services across industry external finance requirements, as none of the interaction coefficients are statistically or economically significant.

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6. Conclusions

The United States has one of the most developed banking systems in the world. Prior research has demonstrated that deregulation, the adoption of lending technology, and securitization have led to improved economic outcomes. However, this paper provides new evidence that, after these improvements, there is significant cross-sectional variation in the effect of information and agency frictions in the banking system, and that these are linked with the internal and external capital market frictions banks face in a given market. To identify this variation, I use oil and gas shale discoveries to obtain exogenous variation in local credit supply to document where and when changes in local credit supply have the largest effect on local firms. If capital were able to flow, absent frictions, to fund positive net present value projects, changes in local credit supply would not affect local firms. Given that changes in local credit supply do affect local firms, it suggests that economically important frictions adversely affect the flow of capital in the banking system.

I find that cross-sectional variation in the effect of changes in credit supply is strongly linked to local bank size. Areas dominated by small banks experience the biggest benefit, in the form of more business establishments in industries with greater external financing requirements, indicating that these lending markets suffer the most from information and agency frictions in the banking system. However, this paper also highlights an important bright side, as other lending markets with a greater presence of large banks do not experience changes in economic activity linked to changes in credit supply. This provides some evidence that the internal capital markets that larger banks have, play a significant role in facilitating economic growth. Alternatively, this paper also highlights that the areas dominated by small banks, which benefit less from deep internal capital markets, do face economically significant lending market frictions, which do affect real outcomes. These results would suggest that additional tools or innovations that could mitigate information or agency frictions for small banks in obtaining funding, may improve outcomes in areas dominated by small banks.

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Endnotes

[1] Prior research outlining the importance of bank internal capital markets for lending includes Houston et al. (1997), Jayaratne and Morgan (2000), Campello (2002), and Gilje et al. (2016). The extent to which this lending translates into an effect on real outcomes depends on frictions that banks face in external funding, the characteristics of a local lending market, and the frictions borrowers face when obtaining loans from financial institutions located outside of their home lending market.

[2] Alternatively, if small local banks have few frictions in accessing external capital markets, all positive NPV projects would be funded prior to the positive deposit shock. In such a case, one would expect a minimal effect on local economic outcomes and for the bank to lend out the new deposit funds in the interbank market or purchase securities.

[3] More broadly, this paper also relates to existing empirical literature focused on the lending and real effects of bank funding shocks, which includes Peek and Rosengren (2000), Petersen and Rajan (2002), Ashcraft (2005), Becker (2007), Khwaja and Mian (2008), Paravisini (2008), Agarwal and Hauswald (2010), Butler and Cornaggia (2011), Chava and Purnanandam (2011), Iyer and Peydro (2011), and Schnabl (2012).

[4] A business establishment is an operating address of a firm; a single firm may have multiple business establishments. I use this as my primary outcome measure as it is among the most granular economic data available at the county-year-industry level during the sample period.

[5] I have excluded all economic outcome measures directly related to oil and gas extraction, construction, real estate, and financial services, because economic outcomes for these industries potentially improve due to reasons unrelated to better local credit supply.

[6] While detailed bank level firm by firm loan data are typically unavailable in the United States, I provide two pieces of evidence that point toward a direct increase in business lending. First, among banks that have all of their branches in a shale boom county, commercial and industrial loans increase after a shale discovery. Second, using Community Reinvestment Act (CRA) data, I find that lending to small businesses increases in shale boom counties from small banks, specifically in boom counties dominated by small banks.

[7] I also provide a variety of robustness tests. Specifically, I show that the results of this study are not driven by (1) preexisting growth trends, (2) any single industry, (3) industry exposure to economic fluctuations as proxied by industry asset beta, (4) changes to local banking markets after a boom, or (5) different small bank size definitions.

[8] Small banks may focus more on relationship lending based on "soft" information relative to transaction lending (Berger and Udell 2006). Sufi (2007) documents that borrowers and lenders are geographically close when information asymmetry is severe.

[9] The size and scale of these payments distinguish these events from other types of economic growth, as well as other types of natural resource extraction (Glaeser et al. 2015, for example).

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[10] $High_j$ is not reported in the regression results because this variable is subsumed by the county-industry fixed effects, $CountyIndustry\ FE_{i,j}$, whereas $Boom_{i,t}$ is not reported because it is absorbed by $CountyYear\ FE_{i,t}$. The high-dimensional fixed effects used for this study are based off of the techniques outlined in Gormley and Matsa (2014).

[11] $High_j$ is not reported in the regression results because this variable is subsumed by the county-industry fixed effects, $CountyIndustry\ FE_{i,j}$, whereas $Boom_{i,t}$, $Small\ Bank_{i,t}$, and $Boom_{i,t} \times Small\ Bank_{i,t}$ are not reported because they are absorbed by $CountyYear\ FE_{i,t}$.

[12] I document that the main results remain statistically significant when using $200 million or $1 billion in assets as the definition of a small bank. The results are also robust to basing this definition on bank holding company assets.

[13] I do not include employment as an outcome variable in my study, because employment data from the county business patterns survey are often suppressed for privacy reasons.

[14] Using three-digit NAICS code industries poses two problems: (1) there are 71 industries as opposed to 14, so there are far fewer comparable Compustat firms for some industries; and (2) there was a change in industry categorization that occurred in 2002–2003, which creates problems when constructing a preboom control period for booms that occur in 2003 and 2004.

[15] The underlying assumption in the Rajan and Zingales (1998) measure is that some industries, for technological reasons, have greater requirements for external financing than others. As Cetorelli and Strahan (2006) highlight, using a measure based on Compustat firms may be considered a cleaner measure, relative to the actual loan amounts small private firms may issue, of the true demand for financing of the firms in the sample. Cetorelli and Strahan (2006) also document a correlation between external finance requirement measures constructed from Compustat and those constructed from the Survey of Small Business Finance.

[16] In Online Appendix A, I also document that high small-bank market share and low small-bank market share experience similar changes in demand, as proxied by retail sales.

[17] Online Appendix C tests for pretrends using a regression specification.

[18] Online Appendix D documents that similar and statistically significant results are obtained when different bank size and holding company definitions are used. Online Appendix E documents that similar and statistically significant results are obtained when holding the banking structure constant as of the year prior to the shale discovery.

[19] Online Appendix F documents that the largest increase in establishments is among establishments with fewer than 10 people, while establishment counts with 10 people or more are unaffected.

[20] Online Appendix B documents the asset betas for each industry and that different asset beta's across high and low external finance requirement industries are not driving the main result. Additionally, Online Appendix B documents that the results are robust to excluding any single industry.

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